Location of Synthetic Aperture Radar Imagery via Range History

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***Abstract*— Geolocation is one of the key synthetic aperture radar (SAR) image processing procedures in SAR-based remote-sensing applications. The conventional SAR imagery geolocation model is based on the range-Doppler (RD) equation, which is derived from the Fresnel approximation of SAR. Currently, there is no other rigorous geolocation model for modern high-resolution SAR; however, this approximation may not be fully applicable for various modern SAR imaging modes and algorithms. This study aims to develop a novel SAR image geolocation model based on the SAR range history (RH) equation, which is derived from the basic SAR backscattered signal property. Moreover, a new geometric calibration model is presented for RH-based locations. Experiments based on real spaceborne SAR data are presented as examples, and the location accuracy of RH and RD location models are compared. The results show that the location accuracies of the RH and RD models are consistent under the existing SAR processing framework, but those for the RH model demonstrate better performance in the azimuth direction. In theory, the proposed RH model can be used for SAR image products generated from complex modes and non-Fresnel-approximation-based imaging algorithms, making it suitable for future SAR systems.**

***Index Terms*— Geometrical calibration, location, range-Doppler (RD), range history (RH), synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active microwave sensor that is capable of continuously monitoring geophysical parameters related to the Earth's surface [1]. One of the key problems of SAR remote-sensing applications is locating SAR imagery, that is, determining the mathematical relationship between the SAR pixel and the real-world target geographic location via SAR imaging geometry [2]. Geolocation is one of the main procedures after the SAR image formation process.

There are two types of models for SAR geolocation: physical and general [3]. Physical models are rigorous and can describe the exact SAR imaging geometry. General models can be regarded as a replacement for physical sensor models to describe various remote-sensing systems in a unified mathematical framework. Thus, the utilization of these two models may be different [4], [5]. In this study, we focused on SAR physical models and spaceborne SAR. Spaceborne SAR is more extensively used in geoinformatic applications, and its imaging geometry is more complex than that of airborne SAR owing to the Earth's surface curvature, the Earth's rotation effect, and the satellite trajectory [6].

The range-Doppler (RD) location model [7] is the only commonly used physical model for SAR geolocation. For the RD location model, the target location can be determined from the intersection point of iso-range lines, iso-Doppler lines, and the Earth's ellipsoid with an instant reference azimuth time. In spaceborne SAR, the RD model utilizes ephemeris data and parameters of the SAR system to produce an estimated image pixel location without reference points and spacecraft attitude knowledge. For practical SAR satellites, such as European Remote Sensing (ERS) [8], ENVISAT Envisat Advanced Synthetic Aperture Radar (ASAR) [9], Advanced Land Observing Satellite (ALOS) [10], COSMO-SkyMed [11], Sentinel, and TerraSAR-X [12], the RD location model is the only practical physical model for SAR geolocation.

The RD location model is derived from the SAR RD imaging model, which assumes separable 2-D processing of the fast-time and slow-time with the deconvolution of chirp signal in the azimuth direction. This assumption also results in separable iso-range and iso-Doppler lines for fast-time and slow-time, respectively, in SAR imaging geometry. This is also called the *Fresnel approximation* of SAR [13]. For more information about the Fresnel approximation in SAR imaging, see Appendix B of this text or [13, Sec. 4.13].

This assumption was proposed early in the development of SAR and may not be fully applicable to modern SAR systems because SAR imaging theories for modern high-resolution SAR have already grown beyond the Fresnel approximation, and other imaging theories without this kind of approximation have already been proposed. For instance, polar format processing is based on the plane wave approximation [13, Sec. 4.11] and is more applicable for spotlight mode SAR systems [14]. SAR imaging theory can also be related to computed tomography [15]. In this way, time-domain processing algorithms such as convolution backprojection [16] and its derivations [17], [18] have been proposed. Moreover, wavefront reconstruction theory [19], which is derived from wave scattering theory [20], can be used for modern SAR without approximation [21].

Considering this, the RD geolocation model may not be directly applied to the postprocessing of the SAR image product when the image is generated from the modern SAR imaging algorithm, which is not derived from the Fresnel approximation because the Doppler parameters may not be

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directly computed during SAR image formation. This limits the application of modern SAR in remote-sensing applications. Thus, a novel SAR target location model is required for modern SAR systems.

To address this problem for the postprocessing of SAR data, this study presents an SAR target location model based on the range history (RH). RH can be obtained from the phase history of the SAR backscattered signal, which is the foundation for all SAR imaging theories. Thus, the RH location model is theoretically applicable for all SAR imaging modes and algorithms. The RD model can be regarded as a first-order approximation of the RH model (see Appendix B for proof). Moreover, similar to the RD model, the new RH model is concise and physically explainable.

The rest of this article is organized as follows. Section II briefly reviews the conventional RD location model and discusses its limitations. Section III presents the proposed RH model and processing details for model solving. Moreover, based on the RH model, a new SAR geometric calibration model is presented. Section IV describes the experiments conducted using real spaceborne SAR data. Spaceborne SAR images from four cities in China are used for geometric calibration and target geolocation experiments. Comparisons are made with the conventional RD location model to validate the proposed RH model. Section V presents the conclusions. Moreover, detailed suggestions for auxiliary data formation using the RH model can be found in Appendix A and the relationship between the RH and RD models can be found in Appendix B.

II. SAR Image Location With the RD Model

After the SAR image formation process, locating the target under a certain Earth coordinate system is normally required to generate higher-level SAR image products [2]. A practical way to find the target location $\boldsymbol{R}_t$ for a given SAR image pixel after the SAR image formation process is to use the intersection point of iso-range lines, iso-Doppler lines, and the Earth's ellipsoid with an instant reference time considering SAR imaging geometry. For simplicity, we suppose that the target is stationary on the ground and located with $\boldsymbol{R}_t$; the RD location equation in the Earth-centered inertial coordinate system can then be expressed as [2], [7]

$$\begin{cases} R(u_0) = ||\boldsymbol{R}_s(u_0) - \boldsymbol{R}_t||_2 \\ f_d = \dfrac{-2(\boldsymbol{V}_s - \boldsymbol{V}_t)\cdot(\boldsymbol{R}_s(u_0) - \boldsymbol{R}_t)}{\lambda R(u_0)} \\ \dfrac{(x_t)^2 + (y_t)^2}{(r_e + h)^2} + \dfrac{\left(z_t^2\right)^2}{r_p^2} = 1 \end{cases} \tag{1}$$

where $\boldsymbol{V}_t$ is the velocity of Earth's rotation, and $R(u_0)$ denotes the instant slant range under reference azimuth time u_0, and $\boldsymbol{R}_s(u_0)$ denotes the instant location for the SAR platform in u_0. Moreover, the target pixel in the image domain with line and sample number can be related with $R(u_0)$ and u_0 using the fast-time and slow-time sampling intervals of SAR, respectively. The imaging geometry with the RD location model is shown in Fig. 1.

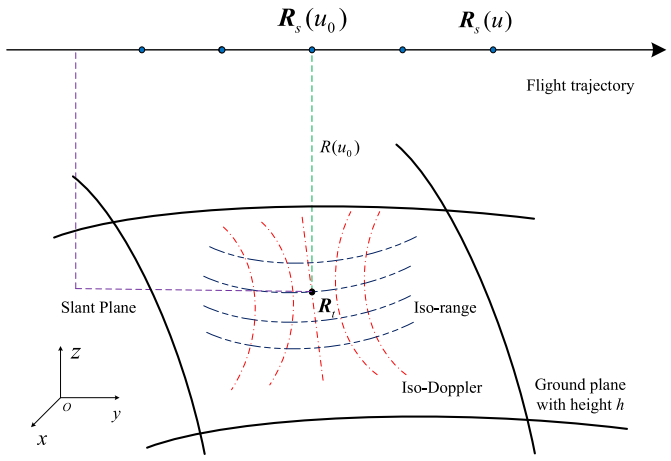


Fig. 1. RD location model where the target $\boldsymbol{R}_t$ can be located using the intersection point of the iso-range line and iso-Doppler line with reference time $u = u_0$ and the known instant platform location $\boldsymbol{R}_s(u_0)$.

The first equation in (1) is the range equation, where $\boldsymbol{R}_t$ denotes the real target location for a given SAR pixel. Moreover, for most SAR imaging algorithms, the pixel number for a given target in its azimuth direction can be related to its reference azimuth time by azimuth compression.

The second equation in (1) is called the Doppler equation, where f_d denotes the Doppler centroid, which can vary with the slant range $R(u_0)$ in spaceborne SAR case [22], and $\boldsymbol{V}_s$ is the instant velocity of the SAR platform under u_0

$$\boldsymbol{V}_s \approx \dot{\boldsymbol{R}}_s(u)|_{u=u_0} \tag{2}$$

where $\dot{\boldsymbol{R}}_s$ denotes the derivative of $\boldsymbol{R}_s$.

The third equation in (1) refers to the Earth ellipsoid equation, where h is the target height, and r_e and r_p are the Earth mean equatorial radius and the polar radius with a flattening factor, respectively. To construct the Earth's ellipsoid equation, the additional digital elevation model (DEM) that contains the target height parameter h can be required. The addition of the Earth ellipsoid equation determines the unique solution for $\boldsymbol{R}_t$ such that the RD location equation is not ill-conditioned.

From the above discussion, it can be seen that the RD location model is concise and mathematically sound; however, the main drawback is that the mathematical foundation of the RD model is based on the RD equations, which are derived from the intersection point of iso-range lines and iso-Doppler lines, as shown in Fig. 1. The parameter f_d may not be directly computed for other modern SAR imaging algorithms. To address this problem, the new location model should consider the basic geometric relationship between the radar and target in the SAR imaging process without further approximation. Additionally, it should be mathematically concise and physically interpretable. In Section III, the RH location model is presented.

III. Proposed RH Model

The mathematical foundation for the SAR target location should be derived from SAR imaging theory in a more general way. In this way, the SAR imaging principle should be

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discussed first to understand the SAR target location. In this section, generic SAR imaging and target location problems are reviewed, and the RH location model is presented.

A. Generic SAR Imaging and Target Location Problems

SAR imaging is a projection that connects 3-D real space and 2-D image space. Normally, a 2-D SAR image space (i.e., the slant plane) is denoted by the range and cross-range directions. For SAR geometric location problems, the key problem is to construct a mathematical relationship between the two spaces after SAR image formation. Moreover, considering different remote-sensing-based applications, the 3-D real space in SAR geolocation can be in a different coordinate system than that in SAR image formation.

Suppose that the SAR imaging geometry can be denoted by Figs. 1 or 2. For a given slow-time u, the radar location is $\boldsymbol{R}_s(u) = [x_s, y_s, z_s]^T$, where ^{T} denotes the matrix transpose and u is omitted for simplicity. The target location is $\boldsymbol{R}_t = [x_t, y_t, z_t]^T$. Then, the RH can be denoted by

$$R(u) = ||\boldsymbol{R}_s(u) - \boldsymbol{R}_t||_2 = \sqrt{(x_s - x_t)^2 + (y_s - y_t)^2 + (z_s - z_t)^2} \quad (3)$$

where $||\cdot||_2$ denotes the vector 2-norm. This RH is a foundation for SAR imaging in modeling the backscattered signal [2], [22]. It should be noted that for the backscattered signal from a specific target, the real RH $R(u)$ between the real locations of the SAR platform and an unknown target can be *unknown*; in this case, we can only construct an estimated RH function for SAR imaging algorithms according to the measured SAR platform parameters, imaging geometry, and imaging modes with a reference point $\boldsymbol{R}_p$ in iso-range circle. We denote this constructed RH in the SAR image formation procedure as $R_d(u)$. However, according to SAR imaging geometry, 3-D real space is projected into the 2-D image space. Then, the reference point $\boldsymbol{R}_p$ may not be exclusive, as shown in Fig. 2, where $\boldsymbol{R}_p$ and its real location $\boldsymbol{R}_t$ are plotted. Moreover, for SAR geolocation purposes, $R_d(u)$ should be reconstructed after SAR image formation.

The choice of $\boldsymbol{R}_p$ varies with different SAR imaging algorithms. For instance, $\boldsymbol{R}_p$ can be set with reference azimuth time with the closest point of the slant range [22]. The intersection point between the iso-range and iso-Doppler lines corresponds to the RD-based target location theory. For time-domain-based algorithms, $\boldsymbol{R}_p$ can be chosen from the given ground grid points. For any $\boldsymbol{R}_p$ that satisfies the SAR imaging geometry, there exists only one point $\boldsymbol{R}'_p$ with $\boldsymbol{R}'_p = \boldsymbol{R}_t$. Thus, the real target location $\boldsymbol{R}_t$ can be determined.

B. Basic RH Location Equation

According to generic SAR imaging theory, SAR imaging can be seen as an inverse problem that utilizes 2-D matched filtering [13], [22]. The impulse response of the SAR system can be constructed using SAR imaging geometry, including the platform information $\boldsymbol{R}_s$ and the reference target point $\boldsymbol{R}_p$. The real target location $\boldsymbol{R}_t$ is projected onto the image space $\boldsymbol{R}_p$, which denotes the set of points shown in Fig. 2 in 3-D space. Considering this principle, the basic RH locating equation can be constructed in the SAR data postprocessing process.

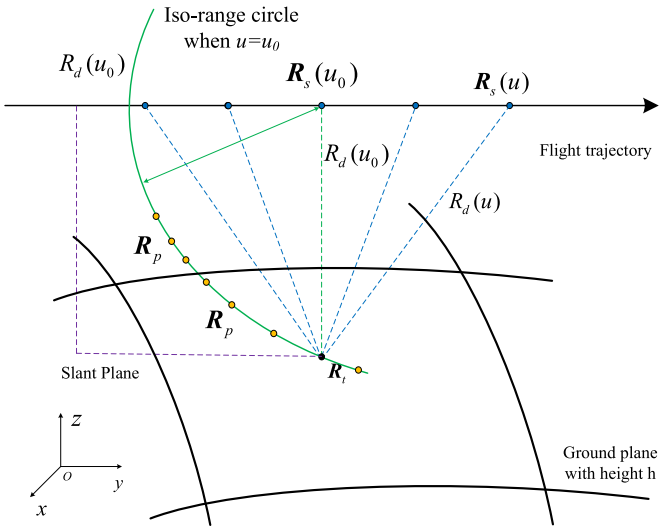


Fig. 2. Proposed RH location geometry where target $\boldsymbol{R}_t$ can be located using the intersection point of the RH function of $R_d(u)$ with multiple u values. The target pixel number in a focused SAR image can be related with $R(u_0)$ and u_0 using the fast-time and slow-time sampling intervals of SAR, respectively. The $\boldsymbol{R}_p$ is the reference point used in SAR image formation and may not be exclusive. However, $\boldsymbol{R}_t$ can be uniquely determined with the additional Earth's ellipsoid function. It should be noted that $\boldsymbol{R}_s(u)$ is the platform trajectory; it may not be a straight line in practical SAR.

After the SAR image formation process, according to SAR imaging geometry, there exists the following RH equation:

$$R_d(u) = ||\boldsymbol{R}'_s(u) - \boldsymbol{R}_p||_2 = ||\boldsymbol{R}_s(u) - \boldsymbol{R}_t||_2 \quad (4)$$

where both $\boldsymbol{R}'_s(u)$ and $\boldsymbol{R}_s(u)$ denote the SAR platform location but $\boldsymbol{R}'_s(u)$ is used for the SAR image formation process and $\boldsymbol{R}_s(u)$ is used for the SAR geolocation process. The two parameters can be in different Earth coordinate systems or can be with different accuracy for different use. However, the platform location $\boldsymbol{R}'_s(u)$ or $\boldsymbol{R}_s(u)$ is always a known parameter that can be recorded by a practical SAR system. This relationship is the foundation for the RH location model. It should be noted that (4) is different from the range equation in (1). The range equation in (1) is an instant slant range when $u = u_0$; however, (4) describes the full RH according to the SAR imaging geometry.

Theoretically, for all available slow-times u, there exists one $\boldsymbol{R}_p$ such that $\boldsymbol{R}_p = \boldsymbol{R}_t$ satisfies (4). However, according to the SAR imaging geometry, for the straight trajectory SAR imaging model (or nearly straight trajectory in real SAR), the solution for (4) may not be unique, as seen in Fig. 2 where all $\boldsymbol{R}_p$ can satisfy this RH function. Details of SAR with a curved trajectory can be found in [23] and [24].

However, it should be noted that the curved trajectory will not violate the proposed RH location model because the platform location vector can be a known parameter for a practical SAR system.

Using the RH equation, the mathematical relationship between $\boldsymbol{R}_p$ and $\boldsymbol{R}_t$ can be constructed using the platform

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trajectory $\boldsymbol{R}_s(u)$, which can be the real platform location in any coordinate system. Moreover, similar to the RD model, to obtain a unique solution of $\boldsymbol{R}_t$, the Earth's ellipsoid function should be included. Thus, the proposed RH model can be expressed as follows.

1) RH equation

$$R_d(u_i) = ||\boldsymbol{R}_s(u_i) - \boldsymbol{R}_t||_2, \quad i = 0, 1, 2, \ldots, K. \tag{5}$$

2) Earth ellipsoid equation

$$\frac{(x_t)^2 + (y_t)^2}{(r_e + h)^2} + \frac{\left(z_t^2\right)^2}{r_p^2} = 1. \tag{6}$$

For the Earth equation, the RH model is similar to the RD model. In this way, to construct the Earth's ellipsoid equation, the additional DEM that contains the target height parameter h can be required. Moreover, for the RH equation (5), it is suggested that $K \geq 2$ considers the SAR imaging geometry. Here, $R_d(u_i)$ denotes the RH used in SAR image construction for different slow-time samples u_i; $R_d(u_i)$ can be constructed using the first equation expressed in (4), which contains reference point $\boldsymbol{R}_p$. Note that the calculation of $R_d(u)$ is performed during the SAR image formation and $R_d(u)$ will be recorded in auxiliary data files by practical SAR imaging systems after a focused SAR image is obtained. It can be a known parameter in SAR geolocating process. Details of how to reconstruct $R_d(u)$ in the SAR geolocation process are discussed below. This is also a critical step during the SAR geolocating process.

Thus, the forward calculation for the RH location model is to calculate $\boldsymbol{R}_t$ from a given SAR image pixel according to imaging parameters, including $\boldsymbol{R}_p$. In addition, the backward calculation for the RH model determines the pixel number (i.e., the closest range R_0 as well as the reference azimuth time u_0) for a given target location $\boldsymbol{R}_t$ from the radar imaging geometry and radar system parameters.

C. Formation of the RH

Although the formation of $R_d(u)$ varies with SAR imaging models and algorithms considering different SAR imaging geometry, the reconstruction of $R_d(u)$ for the RH location equation in the target geolocating process is simple. It should correspond to the exact RH function used for the SAR image formation using the auxiliary information recorded by the SAR imaging system.

For practical SAR systems, including spaceborne and airborne applications, the RH function is normally constructed using the platform location $\boldsymbol{R}_s$, the reference point $\boldsymbol{R}_p$, and other related parameters in the SAR imaging process. After SAR image formation, auxiliary data that contain parameters to reconstruct $R_d(u)$ are recorded and can be found in SAR image product files (normally in the meta files of the image product). Here, we present a method for spaceborne SAR RH reconstruction in the SAR geolocation process using the equivalent squint model as an example.

For spaceborne SAR, considering the Earth's rotation, the Earth's curved surface, and the satellite flight trajectory, the equivalent hyperbolic RH model [22] and equivalent squint RH model [25] are used for spaceborne SAR image formation. The advanced equivalent squint RH model [26] and modified equivalent squint RH model [27] are adopted for modern high-resolution SAR with long synthetic aperture time. A detailed discussion of SAR RH can be found in [28]. Here, for simplicity, we propose the equivalent squint RH equation of spaceborne SAR as an example for the reconstruction of the RH function. If a spaceborne SAR system utilizes the equivalent range model in SAR image formation

$$R_d(u) = \sqrt{R_0^2 + v_0^2 u^2 - 2R_0 v_0 u \cos\phi} \tag{7}$$

and

$$\begin{cases} v_0 = \sqrt{\left(\dfrac{\lambda f_d}{2}\right)^2 - \dfrac{\lambda R_0 f_r}{2}} \\ \phi = \arccos\left(\dfrac{\lambda f_d}{2v_0}\right) \end{cases} \tag{8}$$

where v_0 and ϕ denote the equivalent velocity and the equivalent squint angle, respectively (v_0 is not the platform velocity or the ground velocity of beam-steering in the spaceborne SAR case [22]); λ is the wavelength of the SAR signal; R_0 is the instant slant range when $u = u_0$; and f_d and f_r are the Doppler centroid and Doppler rate for the SAR azimuth signal, respectively. Normally, the range-varying f_d and f_r can be found in SAR image meta files with different forms if this SAR image is generated via Fresnel-approximation-based imaging algorithms. In this way, the RH function $R_d(u)$ can be reconstructed in SAR geolocation problems using the recorded auxiliary information obtained by SAR image formation systems.

Here, another purpose of presenting (7) is to provide the reader with basic ideas for the relationship between the RD and RH models. However, for the actual RH target location model, f_d and f_r are not necessary.

To further illustrate the relationship between SAR image formation and SAR geolocation, we present details of how Doppler parameters are calculated to form the equivalent RH model during spaceborne SAR image formation.

In the SAR image formation procedure, two Doppler parameters are calculated using the reference slow-time u_0 to determine the platform and reference target geometry with

$$\begin{cases} f_d = \dfrac{-2(\boldsymbol{V}_s - \boldsymbol{V}_p) \cdot (\boldsymbol{R}_s - \boldsymbol{R}_p)}{\lambda R} \\ f_r = \dfrac{\left[2\boldsymbol{V}_s \cdot \boldsymbol{V}_s + \boldsymbol{A}_s \cdot (\boldsymbol{R}_s - \boldsymbol{R}_p)\right]}{\lambda R} \end{cases} \tag{9}$$

where $\boldsymbol{V}_s$ is the platform instant velocity, $\boldsymbol{A}_s$ is the instant velocity acceleration, and R denotes the module of the vector $\boldsymbol{R}_s - \boldsymbol{R}_p$. It can be seen that f_d and f_r vary with $\boldsymbol{R}_p$; this is called the range-varying character for spaceborne SAR. This phenomenon can be explained considering spaceborne SAR curved trajectory, the Earth's rotation, and the Earth's curved surface. After SAR image formation, for most spaceborne SAR image products, parameters f_d and f_r for different slant ranges are retained as auxiliary data in the SAR image product meta files (usually ".xml" files) for further postprocessing with the present RD geolocation model. However, for RH location

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models in future SAR systems, f_d and f_r will not be necessary for SAR auxiliary data (see Appendix A).

D. Solution for the RH Model

Here, the solutions for the RH model are presented, including the forward and backward solutions. The forward solution for the RH model is to find $\boldsymbol{R}_t$ for a given SAR pixel, and vice versa. The calculation details are presented using the spaceborne SAR auxiliary data information, which can be easily found in the meta files of any spaceborne SAR image product. The auxiliary data (e.g., sampling rate, pulse repetition frequency (PRF), sampling start time) are records of intermediate parameters during SAR image formation and can be crucial for SAR image postprocessing. The solutions for the RH model are shown below.

1) Forward Solution: The forward solution for the RH model is presented for the spaceborne SAR image product as an example. The problem here is to determine the target location $\boldsymbol{R}_t$ with pixel number (m, n) using the RH model, where m and n denote the pixel number for range and cross-range direction, respectively. The solution details are as follows.

1) Determine the instant slant range $R(u_0)$ with reference azimuth time according to SAR pixel number. Suppose that for a certain SAR image, the nearest sample point in the slant range is $R_{\min}$, the pixel sample number is m, and the instant slant range is
$$R(u_0) = R_{\min} + \frac{cmT_s}{2} \tag{10}$$
where T_s is the fast-time sampling interval. The $R_{\min}$ and T_s can be found in the SAR auxiliary data recorded in a certain SAR imaging mission.
2) Find the reference azimuth time u_0. Suppose that the line number for the target pixel is n, and the reference azimuth time can be calculated as
$$u_0 = u_{\min} + \frac{n}{\mathrm{PRF}} \tag{11}$$
where $u_{\min}$ is the first sample time (the time for the first effective pulse) in SAR imaging, and PRF denotes the equivalent PRF, which can also be found in the SAR image product meta files. For stripmap mode SAR, PRF is the actual PRF in SAR imaging, but for other imaging modes, such as the spotlight mode, PRF denotes the equivalent PRF after slow-time upsampling in the SAR image formation process. In SAR geolocation problems, PRF is a known parameter for this postprocessing step and can be found in SAR image meta files. A similar azimuth upsampling procedure can also be performed for SAR images generated from other modern SAR imaging algorithms [13].
3) Reconstruct the RH function $R_d(u)$. The RH for a given pixel location (m, n) can be reconstructed according to the auxiliary information from the SAR image products. For instance, for spaceborne SAR at present, the auxiliary information about the range varying f_d and f_r can be used to reconstruct the RH according to (7). However, it should be noted that $R_d(u)$ can also be reconstructed without Doppler information. For example, if an SAR image is generated from the time-domain SAR imaging algorithm, the RH can be directly reconstructed using the geometric information of the image grid and SAR trajectory. A similar conclusion can also be made for other SAR imaging models. This should be considered in future SAR missions that use time-domain imaging algorithms [29].
4) Calculate discrete samples of the RH function $R_d(u_0), R_d(u_1), \ldots, R_d(u_K)$ where $K \geq 2$ using the reconstructed RH function $R_d(u)$.
5) Interpolate the platform position vector from ephemeris data or platform inertial measurement unit data and calculate $\boldsymbol{R}_s(u_0), \boldsymbol{R}_s(u_1), \ldots, \boldsymbol{R}_s(u_K)$ for each slow-time sample.
6) Construct the RH location equation using (5) and (6) and then calculate $\boldsymbol{R}_t$. To construct the Earth's ellipsoid equation, the additional DEM which contains the target height parameter h can be required. It should be noted that the RH equation is a typically nonlinear equation that can be solved via iterative methods such as Newton's method with an initial location value set at the imaging scene center. The scene center location information can also be found in current spaceborne SAR image product meta files.

According to the above discussion, for the forward solution for the RH model, the main difference between the RH and RD location models is that the RH model uses the full RH information, whereas the RD model uses the approximated and instant Doppler and slant range information when the reference azimuth time is u_0.

2) Backward Solution: The backward solution for the RH model uses the target real location $\boldsymbol{R}_t$ and platform position vector $\boldsymbol{R}_s$ to calculate the corresponding SAR image pixel location (m, n). Normally, the backward solution for the RH model can be used for SAR system location accuracy verification in range and azimuth direction, respectively.

According to SAR imaging theory, in most SAR imaging algorithms, the target is compressed to the reference azimuth time when performing azimuth compression. In this case, the pixel location (m, n) can be calculated directly only to determine the extreme point for the RH function. Here, we present the backward solution method in detail.

1) Determining the synthetic aperture time according to auxiliary information in SAR image meta files and choose K discrete slow-time samples u_k, where $k = 0, 1, \ldots, K$ where K can be determined according to the model order in the following step 4).
2) Calculate platform position vector $\boldsymbol{R}_s(u_k)$ for each slow-time sample u_k with ephemeris.
3) Construct the discrete RH function $\boldsymbol{R}_s(u_k)$ with the known target location $\boldsymbol{R}_t$.
4) Fit the discrete RH function $\{u_k, R_d(u_k)\}$ using high-order polynomials and find extreme points $[u', R_d(u')]$.

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5) Calculate the corresponding pixel number (m, n) according to the fast-time and slow-time sampling rate using $[u', R_d(u')]$, which corresponds to the slow-time pixel number and the fast-time pixel number.

Notably, the above discussion assumes that the azimuth direction is compressed in the reference azimuth time. This assumption is valid for most frequency-domain-based SAR imaging algorithms. However, for other algorithms (e.g., backprojection), the RH and pixel location can be directly determined via SAR imaging geometry between the platform position and the reference imaging grid. In this way, the backward solution for SAR images generated by time-domain-based imaging algorithms can be more simple.

E. RH Model Calibration

Here, RH model calibration is briefly introduced. SAR geometric calibration theory has been extensively explored for the RD target location model [2]. The main factors that affect SAR target location accuracy include SAR sensor errors, platform errors, the signal propagating factors, terrain-related factors, and processing factors [3]. For practical SAR systems, the final impact of these error factors can be the range and azimuth time error in SAR target location problems [30], [31].

Similar to the geometric calibration theory from the RD model, the geometric calibration for the RH model should also consider those error factors, which brings slow-time error Δ_u for the first slow-time sample time $u_{\min}$ and range error Δ_R for the first slant range $R_{\min}$. Finally, SAR location accuracy is affected by the two errors regardless of the forward or backward solution for RH locating. Considering the range and azimuth time error, the RH-based SAR geometric calibration problem can be

$$\begin{cases} R_d(u_i) = ||\boldsymbol{R}_s(u_i) - \boldsymbol{R}_t||_2, \quad i = 0, 1, 2, \ldots, K \\ \dfrac{(x_t)^2 + (y_t)^2}{(r_e + h)^2} + \dfrac{(z_t^2)^2}{r_p^2} = 1 \\ R_d(u_0) = R_{\min} + \dfrac{cmT_s}{2} + \Delta_R \\ u_0 = u_{\min} + \dfrac{n}{\mathrm{PRF}} + \Delta_u. \end{cases} \tag{12}$$

The solution for this RH geometric calibration model is simple and is similar to the forward solution method for the RH model. However, the parameters to be calculated become Δ_R and Δ_u, whereas $\boldsymbol{R}_s(u_i)$, u_k, and $R_d(u_k)$ can be calculated using SAR auxiliary data using the methods discussed in Section III-C. Thus, there should be one ground control point (GCP) with a precise $\boldsymbol{R}_t$ to determine the slant range error Δ_R and azimuth time error Δ_u. However, the GCP may not be necessary if the above-mentioned error factors can be directly measured, calculated, and compensated, which is similar to RD-based geometric calibration problems [30], [31]. After geometric calibration, the slant range error Δ_R and azimuth time error Δ_u should be compensated in further SAR geolocation problems.

IV. Experiments and Analyses

Experiments were conducted using Chinese high-resolution YG SAR satellite data. This satellite is operated in a sun-synchronous orbit at an altitude of approximately 700 km and has an inclination angle of 97°, spatial resolution within submeter to 20 m, and a swath range from 10 to 200 km. Detailed information about this SAR satellite's geolocation ability and imaging ability can be found in [30]–[33].

The RH and RD target location methods were verified with real SAR image products under each geometric calibration and geolocation scheme. The RH location accuracy was compared with the conventional RD location results in [30]. For comparison, the details of the RD-based geometric calibration and geolocation scheme used in this text can be found in [31]–[33].

Here, the experimental data presented include the high-accuracy corner reflectors mounted on the Songshan Remote Sensing Satellite Calibration Field in China [30] and other manually selected and measured targets within other Chinese areas. Several SAR image products were selected under different imaging conditions. Detailed information on the experimental data can be found in the following discussion.

In the experiments, the RH model was reconstructed using (7) using the auxiliary data in the SAR image product meta files. The SAR imaging algorithm used in this SAR satellite image formation processor was the standard nonlinear chirp scaling (NCS) algorithm [34], in which range-varying Doppler information for f_d and f_r is stored in the auxiliary data files. In this way, both the RD and RH location models can be used for geometric location verification experiments. Moreover, using (7), the RH function could be easily reconstructed in SAR geolocation problems. The experiments were conducted as follows.

1) Corner reflector location experiments with Songshan area SAR image products. Corner reflectors mounted on the Songshan Remote Sensing Satellite Calibration Field in China have known geolocation with high accuracy. The accuracy of the corner reflectors is within meters.
2) Location experiments with three cities in China, for which targets were manually selected and measured. SAR images were selected under different imaging conditions to verify the proposed RH target position method.

In the following text, detailed experiments are presented, and we prove that the proposed RH model can replace the conventional RD location model with a high location accuracy under the existing SAR geolocation scheme. According to Appendix B, the conventional RD model can be regarded as a first-order approximation for the RH model with reference slow-time u_0.

A. Experimental Data

Here, detailed information regarding the experimental data is presented. SAR images were selected using sliding spotlight mode data products. The sliding spotlight mode is one of the main imaging modes for this satellite and is a hybrid imaging mode between the stripmap and spotlight modes [35]. The image scene width and resolution can be properly controlled using the sliding spotlight parameter. This imaging mode is more flexible than spotlight mode and stripmap mode SAR, but the RH model can be more complex.

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TABLE I

SAR IMAGE INFORMATION FOR THE SONGSHAN AREA, WHERE SS DENOTES SLIDING SPOTLIGHT MODE. IN EACH SAR IMAGE, ONE CORNER REFLECTOR IS SET AS THE GCP FOR GEOMETRIC CALIBRATION, AND OTHER CORNER REFLECTORS ARE SET AS CPS FOR LOCATION ACCURACY VERIFICATION

Image number	Imaging time	mode	Range × Azimuth	pulse-width&bandwidth	GCP/CP
01	12/11/2015	SS	6.3 km × 9.7 km	29.2 us & 512 MHz	1/3
02	12/13/2015	SS	3.1 km × 9.3 km	26.2 us & 600 MHz	1/3
03	12/30/2015	SS	3.1 km × 9.3 km	26.2 us & 600 MHz	1/3
04	12/31/2015	SS	4.1 km × 10.0 km	29.2 us & 600 MHz	1/3
05	01/05/2016	SS	3.6 km × 9.5 km	26.2 us & 600 MHz	1/3
06	01/10/2016	SS	4.6 km × 10.1 km	29.2 us & 600 MHz	1/3
07	03/31/2016	SS	9.5 km × 26.7 km	24 us & 256 MHz	1/4

TABLE II

SAR IMAGE INFORMATION FOR EXPERIMENTS BASED ON MANUALLY SELECTED TARGETS. IN EACH SAR IMAGE, ONE GROUND TARGET IS SET AS THE GCP FOR GEOMETRIC CALIBRATION AND OTHER GROUND TARGETS ARE SET AS CPS FOR LOCATION ACCURACY VERIFICATION

City	Image number	Imaging time	Mode	Range × Azimuth	pulse-width&bandwidth	GCP/CP
Nanjing	01	05/20/2017	SS	6.1 km × 12.0 km	29.2 us & 512 MHz	1/13
	02	05/21/2017	SS	6.1 km × 11.2 km	29.2 us & 512 MHz	1/13
	03	05/26/2017	SS	5.7 km × 11.5 km	26.2 us & 600 MHz	1/16
	04	06/05/2017	SS	4.8 km × 10.4 km	26.2 us & 600 MHz	1/11
	05	06/13/2017	SS	4.8 km × 10.4 km	26.2 us & 600 MHz	1/14
	06	06/14/2017	SS	6.6 km × 11.1 km	29.2 us & 512 MHz	1/13
Ningbo	01	01/10/2016	SS	6.3 km × 9.7 km	29.2 us & 512 MHz	1/19
	02	01/11/2016	SS	6.8 km × 12.7 km	29.2 us & 512 MHz	1/19
Shanghai	01	06/26/2017	SS	5.4 km × 11.2 km	29.2 us & 600 MHz	1/18
	02	07/04/2017	SS	6.1 km × 9.8 km	29.2 us & 512 MHz	1/18

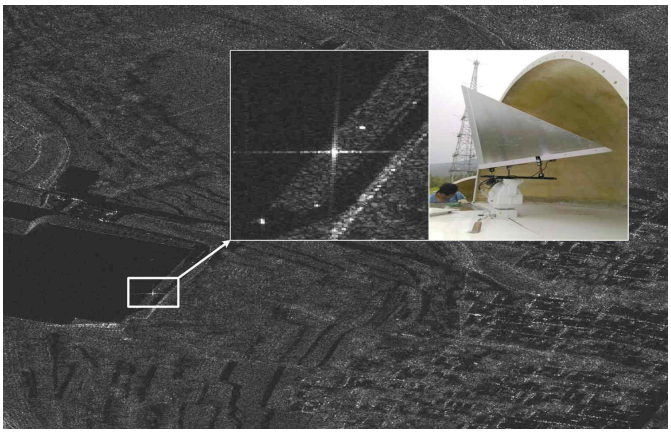


Fig. 3. Segmented SAR image for one of the SAR images in the Songshan experiment. A zoomed-in image of the corner reflector is shown, along with an optical photograph of the corner reflector.

In experiments based on the corner reflectors of the China Songshan Remote Sensing Satellite Calibration Field, the corner reflectors were mounted within the Songshan area with high (submeter) accuracy. The SAR image information is shown in Table I for the Songshan area experiment. It can be seen that seven SAR images with different time and bandwidth products are selected; thus, the target location accuracy is examined with different imaging parameters. From practical measurement, the position accuracy for the corner reflectors is within 0.3 m.

Fig. 3 shows one of the Songshan SAR images used in this experiment, along with a zoomed-in sinc-like subimage for the corner reflector. An optical photograph for this corner reflector is presented in Fig. 3. The SAR image is well-focused, and the corner reflector is set in an appropriate direction so that a 2-D sinc-like function character can be seen from the SAR image.

Here, another experiment based on manually selected targets was performed. Targets were selected based on distinct features in SAR images (e.g., crossroads that can be easily distinguished in SAR images). All of the actual locations of the targets were manually measured.

In this location experiment, SAR images for three cities in China were selected with different imaging conditions, as shown in Table II. Ten SAR images were selected for this experiment. For each SAR image, targets with known pixel numbers and measured location were divided into two groups: GCPs and check points (CPs). One GCP was selected for geometric calibration for each SAR image, and then other targets were set as CPs to verify the location accuracy. The imaging parameters for selected SAR images were different; thus, the target location accuracy for the proposed RH model could be verified.

Fig. 4 presents part of the SAR imagery for Shanghai city, where one of the targets in this image is presented with a zoomed-in image. Moreover, the optical photograph for this target is also shown in Fig. 4 where the red cross in the image shows the real location of this target.

The details of the location experiments are provided in Sections IV-B and IV-C. To evaluate the location accuracy for the given CPs in the range and azimuth direction, a backward solution was performed.

B. Songshan Corner Reflector Experiment: Results and Discussion

Here, the results of corner reflector experiments of the Songshan Remote Sensing Satellite Calibration Field in China

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TABLE III

STATISTICAL RESULTS FOR CORNER REFLECTORS OF THE SONGSHAN REMOTE-SENSING SATELLITE CALIBRATION FIELD IN CHINA FOR TARGET LOCATION ERRORS BEFORE AND AFTER RECALIBRATION FOR THE RD AND RH MODEL IS PRESENTED. THE RESULTS ARE COMPARED WITH THE RMSES

Image	Method	Accuracy **after** calibration (RMSE)			Accuracy **before** calibration (RMSE)		
		Range/m	Cross-range/m	Plane/m	Range/m	Cross-range/m	Plane/m
01	RD	0.20	0.61	0.65	16.23	335.02	335.41
	RH	**0.20**	**0.34**	**0.39**	**16.21**	**167.85**	**168.63**
02	RD	0.38	0.68	0.77	16.63	208.84	209.50
	RH	**0.38**	**0.32**	**0.50**	**16.63**	**104.57**	**105.88**
03	RD	0.35	2.43	2.46	17.32	65.02	67.28
	RH	**0.35**	**1.28**	**1.33**	**17.31**	**32.61**	**36.92**
04	RD	0.42	0.55	0.69	18.04	304.78	305.31
	RH	**0.42**	**0.24**	**0.48**	**18.02**	**152.48**	**153.54**
05	RD	0.11	0.83	0.84	0.07	133.14	133.14
	RH	**0.11**	**0.39**	**0.40**	**0.07**	**66.78**	**66.78**
06	RD	0.11	0.64	0.65	18.24	172.84	173.80
	RH	**0.11**	**0.33**	**0.35**	**18.23**	**86.43**	**88.33**
07	RD	0.20	0.64	0.67	16.65	324.39	290.52
	RH	**0.20**	**0.33**	**0.38**	**16.64**	**162.30**	**145.92**



Fig. 4. Segmented SAR image for Shanghai city with the target shown in the zoomed-in optical photograph. The location measurement scenario is illustrated; the red cross denotes the actual location of the target.

are presented. The target location accuracy was evaluated after geometric calibration for each SAR image, and the RH and RD models were compared.

The corner reflectors in SAR images were divided into two groups: GCPs and CPs. For geometric calibration problems, one GCP was used to calculate the range and azimuth error parameters in (12) for each SAR image. Moreover, CPs in each SAR image were used for the target location accuracy evaluation after geometric calibration. In this experiment, the precise ephemeris is used to determine the satellite location vector $\boldsymbol{R}_s(u)$ for a given slow-time u. The related parameters for the calculation of the RH and RD location models can all be found in the SAR image product meta files.

To evaluate the recalibration performance of both the RD and RH models, the target location accuracy for both methods before geometric calibration is presented in Fig. 5(a). For RH geometric calibration, (12) was used to determine the range and azimuth time error. For the RD geometric calibration, the method in [30] was used. The target location accuracy after geometric calibration is shown in Fig. 5(b).

In both figures, each point denotes a CP's location error, and the number in the figures denotes the SAR image number. Different shapes for the point also denote different SAR image numbers. For comparison, location errors for the RD and RH models are presented in different colors. For example, the "01 RD" shows the location error for the first SAR image with different CPs. Results from Fig. 5(a) and (b) show the effectiveness of the geometric calibration for both the RD and RH location models.

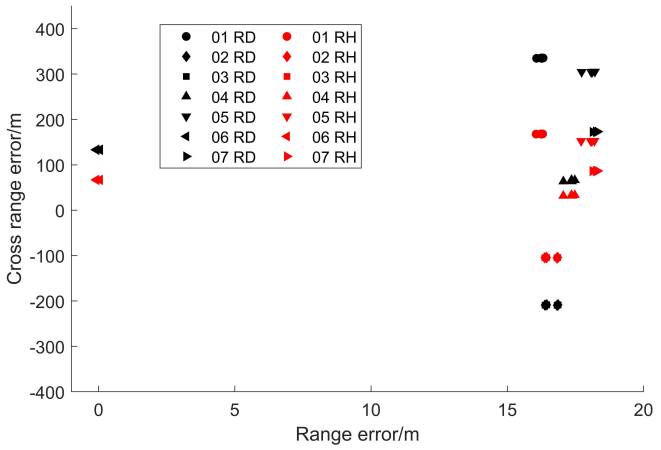
A quantitative comparison of Fig. 5 is made in Table III where the statistical results for target location error are presented. As shown in Table III, the target location accuracy increases drastically after geometric calibration for both the RD and RH models. The root mean square errors (RMSEs) for the range and azimuth are within 1 m; moreover, the distribution for the geometrically calibrated location error shows more evident random error. The results validate the geometric calibration for both the RD and RH models.

For target location accuracy before geometric calibration, Fig. 5(a) shows that the error distribution has some regular features. The mean value for range errors lies tens of meters in this direction, which results from the slant range errors caused by error factors such as signal propagation delay and sample time error in the range direction. Furthermore, distributions of the azimuth error vary between SAR images. The mean value for azimuth location errors is hundreds of meters for each CP. This results mostly from the azimuth time error of the SAR system.

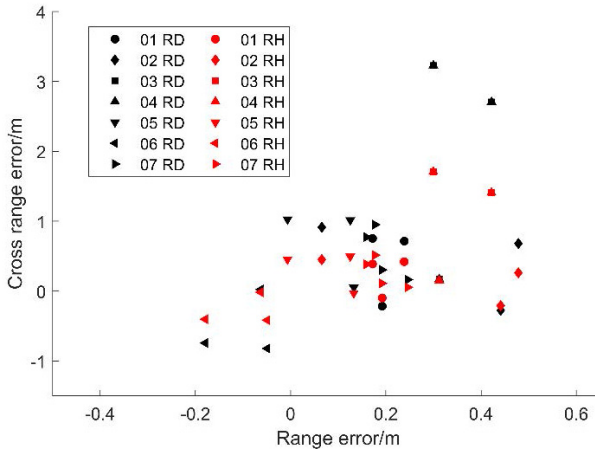
After the geometric calibration, the results in Fig. 5(b) and Table III reveal that the geolocation accuracies for the RH and RD models are mostly consistent in the range direction. However, from the geolocation results in the azimuthal direction, the RH model is more accurate than the RD model. The reasons for this phenomenon are discussed below.

Theoretically, from discussions of the RH model in Section III, the RH and RD models should be equivalent. However, for the conventional RD model used in practice, the first-order coefficient for RH (i.e., f_d) is used, and this only considers the range-varying character and ignores the

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(a)



(b)

Fig. 5. Target location accuracy comparisons between the RD and RH models before and after geometric calibration. For both, each point denotes one control point location error, while the numbers denote the SAR image number. Different shapes for the point also denote different SAR image numbers. For comparison, location errors for the RD and RH models are presented in different colors. For example, the "01 RD" denotes the location error for the first SAR image with different CPs. Location error (a) before and (b) after geometric calibration.

azimuth-varying character for f_d. In this way, the f_d parameter used in RD location is only an approximation when $u = u_0$ with $\boldsymbol{R}_s(u_0)$. Thus, the Doppler function in (1) was constructed with a reference azimuth time u_0. However, for practical high-resolution SAR imaging systems, coefficients for RH can be both range- and azimuth-varying (i.e., f_d and f_r or other coefficients vary with range R and azimuth time u).

However, for the proposed RH model, the RH can be directly reconstructed using the SAR platform location vector $\boldsymbol{R}_s$ with different u and the target location vector $\boldsymbol{R}_t$ directly without approximation. The solution details for the RH model are provided in Section III.3.2. Thus, the RH location model shows better azimuth location accuracy compared with the RD model in real SAR data processing.

C. Manually Selected Target Experiments: Results and Discussion

Here, the results of manually selected target experiments are presented. In these experiments, the targets were individually measured. Similar to the Songshan experiment with corner reflectors, the targets were divided into two groups: one GCP in each SAR image was used for geo-calibration, and other targets in each SAR image were set as CPs for location accuracy verification. It should be noted that the target location accuracy for these manually selected targets may be less precise than that of corner reflectors because of human error in picking the targets for each SAR image. However, the number of CPs was larger than that in the Songshan experiments.

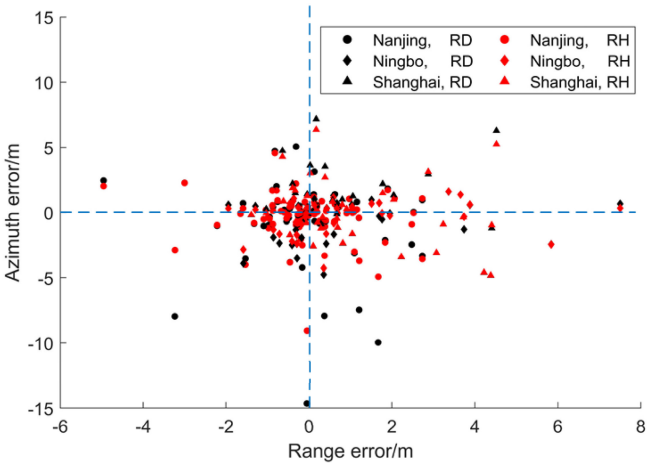


Fig. 6. Location error distribution after geometric calibration for both the RD and RH models. Points with different shapes denote the geometric location error in range and azimuth direction.

TABLE IV

STATISTICAL RESULTS FOR LOCATION ACCURACY FOR DIFFERENT CITIES IN CHINA WITH MANUALLY SELECTED TARGETS UNDER THE RD AND RH LOCATION MODELS, WHERE μ_r AND σ_r DENOTE THE RANGE ERROR MEAN VALUE AND STANDARD DEVIATION, RESPECTIVELY; AND μ_a AND σ_a DENOTE THE AZIMUTH ERROR MEAN VALUE AND STANDARD DEVIATION, RESPECTIVELY

City	Statistics	RD/m	RH/m
Nanjing	μ_r	-0.07	**-0.07**
	σ_r	1.25	**1.25**
	μ_a	-0.63	**-0.40**
	σ_a	2.83	**1.79**
Ningbo	μ_r	0.86	**0.86**
	σ_r	2.00	**2.00**
	μ_a	-0.55	**-0.33**
	σ_a	1.64	**1.31**
Shanghai	μ_r	1.00	**1.00**
	σ_r	1.67	**1.67**
	μ_a	0.30	**0.12**
	σ_a	2.59	**2.38**

For simplicity, the target location error distribution after geometric calibration is shown in Fig. 6, where the points denote the location error distribution for a given CP.

From Fig. 6, it can be seen that the location error distribution after geo-calibration shows some random character. The mean value for the range and azimuth error was close to 0 m for both the RD and RH models. Moreover, the error distribution shows randomness, which means that only random measuring errors remain after geo-calibration. The results validate the geometric calibration method for both the RD and RH models. However,

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from Fig. 6, the error distribution is mostly influenced by the azimuth direction. This is because of errors introduced by the manually selected CPs. Moreover, the geolocation results for the RH model were more accurate than those of the RD model in the azimuth direction.

Table IV lists the statistical results for the target location errors where the mean value u and the standard deviation σ are calculated individually for range and azimuth direction. It can be seen that similar to the previous Songshan experiment, the azimuth location error for the RH model was better than that for the RD model. However, in terms of range error, accuracies for RD and RH were similar. The reason for this phenomenon was discussed in relation to the Songshan experiment. Thus, the performance of the proposed RH model has been systematically validated.

V. Conclusion

In this study, a novel SAR location model based on the RH was presented. The proposed RH locating model connects SAR imaging geometry in SAR image formation and SAR geolocating processes using the RH function $R_d(u)$. The RH model was based on basic SAR imaging theory without approximations and can be more applicable for modern SAR image products generated by various imaging modes and algorithms. The conventional RD model can be regarded as a first-order approximation of the proposed RH model. Thus, Doppler information may not be necessary according to the RH model construction and solution. Moreover, the geometric calibration model was also presented in the RH model. From the results of the real spaceborne SAR data, we note the following.

1) Both the RH and RD models demonstrated acceptable location results in the proposed experiments, thereby validating the proposed RH location model. Thus, the newly introduced RH model is compatible with the existing SAR image product processing framework under the RD location model.
2) The location accuracy for the azimuth direction was better for the RH model than the RD model in the practical data processing.

For brevity, the geometric calibration experiments presented here used one GCP to calculate the range and azimuth errors Δ_R and Δ_u for calibration. However, for most geometric calibration topics without GCPs, including atmospheric correction, system delay corrections can be easily found in the literature on the classical RD location model [30], [31], and the conclusions are similar. Future studies may also include more specific geometric calibration factors in the RH calibration problems, such as SAR "stop-go" approximation and wave propagation delay by the atmosphere.

It should be pointed out that although our discussion is based on spaceborne SAR, the construction and solution of the proposed RH model are not limited to this case. The key issue for RH locating model in SAR geolocating process is to reconstruct the same RH function that is used and recorded in the SAR image formation process. Hence, for other platform-based-SAR approaches (e.g., airborne SAR), the construction of the RH model is similar to that of the spaceborne SAR case with real-time $\boldsymbol{R}_s$ measured by the airplane's inertial measurement unit. Moreover, considering the motion errors, the construction of the RH $R_d(u)$ should also correspond to the RH model with motion error which is calculated and recorded by the airborne SAR image formation algorithm and the corresponding autofocus algorithm [36].

Further research on this topic should also include image product processing for RH models such as block adjustment or interferometric processing, where target location information should be determined. In this way, a more applicable SAR location model can be used for future SAR systems.

Appendix A
Suggestions for SAR Auxiliary Data Construction Using the RH Model

The RH model can be used directly to form SAR geolocation-based products with the proper auxiliary data saved in SAR imaging. In Table V, we present suggestions for SAR auxiliary data construction for future SAR systems using the RH model. The auxiliary data used in the RD location are also presented for comparison.

Appendix B
Relationship Between the RH and RD Models

The relationship between the RH and RD location models is mathematically presented. The RD location equation in (1) contains three types of equations, including the range, Doppler, and Earth equations. However, the RH model contains only two types of equations: the RH (5) and the Earth equation (6). The following discussion proves that the RD location model uses a first-order approximation coefficient of the RH to construct the Doppler equation.

Suppose that the platform location $\boldsymbol{R}_s$ varies with u, and the velocity $\boldsymbol{V}_s$ can be approximated with

$$\boldsymbol{V}_s \approx \boldsymbol{V}_s(u_0) = \dot{\boldsymbol{R}}_s(u_0) \tag{B.1}$$

and the Doppler parameters f_d with reference azimuth time u_0 can be constructed by

$$\begin{aligned} f_d &= \frac{-2}{\lambda R(u_0)} \boldsymbol{V}_s \cdot (\boldsymbol{R}_s(u_0) - \boldsymbol{R}_t) \\ &= \frac{-2}{\lambda R(u_0)} \boldsymbol{V}_s \cdot (\boldsymbol{R}_s(u_0) - \boldsymbol{R}_p). \end{aligned} \tag{B.2}$$

It should be noted that for (B.2), in SAR imaging, f_d is calculated via $\boldsymbol{R}_p$ (here, we use $\boldsymbol{R}_p$ to denote the reference points used in SAR imaging geometry construction and use $\boldsymbol{R}_t$ to denote the real target location). In SAR location problems, $\boldsymbol{R}_p$ and f_d are known, but the real target location $\boldsymbol{R}_t$ is unknown. Thus, $\boldsymbol{R}_t$ can be found using target location models such that $\boldsymbol{R}_t \in \{\boldsymbol{R}_p\}$. This is the foundation of the following discussion.

Next, the relationship between the RH and RD models is constructed. Here, the main problem in SAR target location is to use f_d or $\boldsymbol{R}_p$ to find $\boldsymbol{R}_t$. In the RH model, suppose that $F(x_s, y_s, z_s)$ denotes the RH (4) for the $\boldsymbol{R}_p$-related function, where $\boldsymbol{R}_s(u) = [x_s, y_s, z_s]^T$. The $\boldsymbol{R}_t$ denotes the real target

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TABLE V

SAR AUXILIARY DATA SUGGESTIONS FOR RH AND RD MODEL GEOLOCATION APPLICATIONS. THREE AUXILIARY DATA FORMATIONS ARE LISTED AS SUGGESTIONS FOR THE RH-BASED SAR DATA PRODUCT. THE FIRST FORMATION FOR RH IS COMPATIBLE WITH THE EXISTING RD-BASED SAR GEOLOCATION PROCESSING SCHEME BUT DOES NOT SUPPORT TIME-DOMAIN-BASED IMAGING ALGORITHMS. THE PARAMETERS IN THIS TABLE ARE AS FOLLOWS: PRF, EQUIVALENT PRF; F_s, FAST-TIME SAMPLING RATE; $u_{0\ \min}$, TIME FOR THE FIRST PULSE; $R_{0\ \min}$ SLANT RANGE FOR THE FIRST RANGE BIN; f_d, DOPPLER CENTROID; f_r, DOPPLER RATE; v_0, EQUIVALENT VELOCITY; ϕ, EQUIVALENT SQUINT ANGLE; h, TARGET HEIGHT; R_0, CLOSEST SLANT RANGE FOR THE TARGET; u, SLOW-TIME; AND $\boldsymbol{R}_p$, REFERENCE POINT

Location model	Auxiliary data for geolocation under different imaging algorithms	
	Frequency domain based	Time domain based
RD	PRF, F_s, u_{0min}, R_{0min} f_d, $\boldsymbol{R}_s(u)$, h	Do not support
RH	(1) PRF, F_s, u_{0min}, R_{0min}, $\boldsymbol{R}_s(u)$, h, f_d, f_r	
	(2) PRF, F_s, u_{0min}, R_{0min}, $\boldsymbol{R}_s(u)$, h, v_0, ϕ	
	(3) PRF, F_s, u_{0min}, R_{0min}, $\boldsymbol{R}_s(u)$, h, Samples for $R(u)$ with different R_0 and u	
	(4) PRF, F_s, u_{0min}, R_{0min}, $\boldsymbol{R}_s(u)$, h, Polynomial coefficients for constructing $R(u)$ with R_0 and u	
	(5) PRF, F_s, u_{0min}, R_{0min}, $\boldsymbol{R}_s(u)$, h, Imaging ground plane grid samples for $\boldsymbol{R}_p$	

location, and $\boldsymbol{R}_p$ denotes all points that satisfy the SAR imaging geometry. For simplicity, u is omitted. The derivative of $F(\cdot)$ is

$$\frac{d}{du}||\boldsymbol{R}_s - \boldsymbol{R}_p||_2 = \frac{\partial F}{\partial x_s}\frac{dx_s}{du} + \frac{\partial F}{\partial y_s}\frac{dy_s}{du} + \frac{\partial F}{\partial z_s}\frac{dz_s}{du}. \quad \text{(B.3)}$$

The above equation can also be denoted with

$$\begin{cases} \dfrac{\partial F}{\partial x_s} = \dfrac{(x_s - x_p)}{\sqrt{(x_s - x_p)^2 + (y_s - y_p)^2 + (z_s - z_p)^2}} \\ \dfrac{\partial F}{\partial y_s} = \dfrac{(y_s - y_p)}{\sqrt{(x_s - x_p)^2 + (y_s - y_p)^2 + (z_s - z_p)^2}} \\ \dfrac{\partial F}{\partial z_s} = \dfrac{(z_s - z_p)}{\sqrt{(x_s - x_p)^2 + (y_s - y_p)^2 + (z_s - z_p)^2}}. \end{cases} \quad \text{(B.4)}$$

The platform velocity $\boldsymbol{V}_s$ can be expressed as $\boldsymbol{V}_s = [v_x, v_y, v_z]$, such that

$$\boldsymbol{V}_s = \left(\frac{dx_s}{du}, \frac{dy_s}{du}, \frac{dz_s}{du}\right). \quad \text{(B.5)}$$

Using (3) and (B.3), the derivative can be expressed as

$$\frac{d}{du}||\boldsymbol{R}_s - \boldsymbol{R}_p||_2 = \boldsymbol{V}_s\frac{(\boldsymbol{R}_s - \boldsymbol{R}_p)}{R}. \quad \text{(B.6)}$$

It can be seen that using (B.6) with coefficient $[-2/(\lambda R(u_0))]$, the Doppler centroid f_d can be constructed for the RD model. In fact, (B.6) corresponds to (B.2) with the $\boldsymbol{R}_p$ term. The SAR location model is then used to calculate $\boldsymbol{R}_t$. Thus, it can be concluded that for the RD model, the Doppler equation is from the first-order approximation coefficient of the RH function with reference azimuth time u. Moreover, in spaceborne SAR, f_d also varies with the slant range.

It should be noted that the Taylor expansion for (B.3) with a second-order approximation is the well-known Fresnel approximation. The second term of this expansion of RH function results in a chirp signal characteristic for SAR backscattered signal in the azimuth direction [13].

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